

OMID VAHEB

📍 Toronto ✉ ovaheb@gmail.com 🔗 LinkedIn 🌐 GitHub 📁 Portfolio 🎓 Publications

Professional Summary

- **ML Engineer** with 4+ years of industrial & academic experience advancing **LLMs**, **Multimodal AI**, and **Large-scale Model Optimization** across REDspace, Vector Institute, NRC, and UofT.
- Expert in **PyTorch**, **TensorFlow**, **C++**, and **Cloud platforms** (AWS, GCP), with a proven track record deploying **generative**, **multimodal**, and **multi-agent** AI systems into production.

Professional Experience

Machine Learning Engineer | [REDspace](#) 📍 Remote, Canada | 📅 Jan. 2025 – Present

- Boosted **Automatic Speech Recognition** accuracy by cutting **Hallucinations** and **Word Error Rate** by 50%, reducing operational costs, and confirming improvements via **A/B testing**.
- Deployed generative error correction module into Mockingbird Dubbing Solution. Engineered a **multimodal** PoC using **OpenAI** and **open-source** models, achieving SOTA accuracy.

Machine Learning Associate | [Vector Institute](#) 📍 Toronto, Canada | 📅 Sep. 2024 – Present

- Developed **multi-agent systems** for startups & medium enterprises, and collaborated with cross-functional teams to integrate effective AI solutions.
- Designed chatbots utilizing **RAG** and **post-training** techniques, such as **LLM fine-tuning**.

Applied Scientist Intern | [National Research Council Canada](#) 📍 Remote, Canada | 📅 Jun. 2023 – Sep. 2024

- Reduced **MAE by 15%** in astronomical **image reconstruction** by optimizing Restormer **Transformer** and U-Net architecture. Implemented in **PyTorch** and deployed on a **distributed multi-GPU** cluster.
- Generated **large-scale** synthetic data from real galaxy morphologies, reducing CPU **synthesis time by 20%**. Developed preprocessing pipelines for raw telescope data.

Research Assistant | [University of Toronto](#) 📍 Toronto, Canada | 📅 Sep. 2022 – Sep. 2024

- Improved astronomical **image denoising** by **15dB PSNR**, boosting **object detection by 7%**, and cutting observation costs **by 66%**. Developed an **unsupervised** training framework mixing N2N and SURE loss.
- Optimized data loaders, reducing memory usage **by 80%**, execution time **by 60%**, and limiting disk access, **accelerating model training**.

Research Assistant | [Computational Audio-Vision Lab](#) 📍 Remote | 📅 Apr. 2021 – Aug. 2022

- Achieved **92% accuracy** in autism detection from infants' crying voice by fine-tuning Google's YAMNet model and **ensemble** classifier using **TensorFlow**.
- Attained **99.5%** similarity & consistency with human annotations by **fine-tuning** YAMNet and using energy-thresholding to **preprocess** voice recordings.

Data Scientist | [Virasad Startup](#) 📍 Remote | 📅 May 2020 – Apr. 2021

- Reduced maintenance costs **by 10%** with real-time **anomaly detection** using LSTM and ARIMA for production line time series forecasting.

Related Skills

- **Programming and Data:** Python, C/C++, Linux Shell, Javascript, MATLAB, SQL (MySQL), R
- **Machine Learning & AI:** PyTorch, TensorFlow, Hugging Face, Langchain, PydanticAI, LlamaIndex, JAX, Fastai, AutoML, NLTK, OpenCV, OpenAI API
- **Deployment & Cloud:** AWS (Sagemaker), GCP (Vertex AI, BigQuery), Docker, Streamlit, Wandb, Git

Education

Master of Applied Science in Electrical and Computer Engineering 2022 – 2024
University of Toronto; GPA: 3.86/4 *Toronto, Canada*

B.Sc. in Electrical Engineering, Minor in Computer Engineering 2017 – 2022
University of Tehran; GPA: 3.92/4 *Tehran, Iran*